

TAL GOTTESMAN

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EDUCATION

Université Paris cité PhD in pure mathematics Under the supervision of Baptiste Rognerud. Subject : Fractionally Calabi–Yau Lattices IMJ-PRG Full funding from École doctorale Sciences Mathématiques Paris centre.	<i>September 2021 - December 2024</i>
Université Paris cité Masters of pure and applied Mathematics UFR de Mathématiques	<i>September 2019 - June 2021</i> with highest honors
University of Cambridge Master of advanced study in Pure Mathematics Department of Pure Mathematics and Mathematical Statistics	<i>October 2018 - June 2019</i> Pass
Sorbonne Université (former UPMC) Double major Bachelor Mathematics and Physics Faculté des Sciences	<i>September 2015 - July 2018</i> with highest honors

WORK EXPERIENCE

Ruhr Universität Bochum <i>Post-doc</i> · In Prof. Dr. Karin Baur’s team · Four hours weekly teaching duties	October 2024 - present
Université Paris Cité <i>Complementary teaching duty as a phd student</i> · Four hours weekly teaching duties	September 2021 - August 2024
Sorbonne Université <i>Tutoring</i> · Tutored 15 first year undergraduate student in mathematics in weekly meetings.	January 2018 - May 2018
Private Lessons · High School level mathematics. Weekly	2016 and 2019/2021

PAPERS

Publications

- T. Gottesman, *Antichains in the representation theory of finite lattices*, Séminaire Lotharingien de Combinatoire (2024).

- T. Gottesman, *Fractionally Calabi-Yau lattices that tilt to higher Auslander algebras of type A*. Advances in mathematics. <https://arxiv.org/abs/2406.09148>

Preprints

- T. Gottesman, Viktória Klász, Markus Kleinau, Rene Marczinzik, *Pure minimal injective resolutions and perfect modules for lattices*. <https://arxiv.org/abs/2511.03385>.

In preparation

- J. Garber, L. Goltermann, D. Horiatakis, D. König, T. Gottesman, *Counting Boolean antichains*.

SUPERVISING

Summer projects:

- Lucas Pouillart, *Root systems and the word problem in finite coxeter groups*, summer 2023.
- J. Garber, L. Goltermann, D. Horiatakis and D. König, *Counting Boolean antichains*, summer 2025, Dive into Research funded by Combinatorial synergies.

Masters students:

- Julie Garber, *Combinatorial contructions in the representation theory of finite posets*, Winter semester 2025-2026. University of Bochum.

ORGANISING AND OTHER RESPONSABILITIES

Diversity Officer of the Faculty of mathematics 2025-ongoing
university of Bochum

- I followed a faculty organised training afternoon on how to welcome students and faculty workers that face discriminations.

Learning Seminar on Cohen-Macaulay posets Winter semester 2025-2026
University of Bochum

- I organised this seminar which was a joint seminar between the team of Prof.Dr. Karin Baur and the team of Prof. Dr. Markus Reineke.

Collaborations in Algebra, Representation theory and Ethics October 2025
ENS Lyon

- Conference with 70 participants, 18 math research talks and two minicourses, 4 talks on ethical topics followed by dicussions.
- Together with my coorganisers Alice Bouillet, Luca Casarin, Luca Francone, and Matilde Maccan, I among other things, contacted funding bodies, designed the poster of the event (with Luca Casarin), maintained the website, invited speakers.

Phd student representative September 2022 - September 2024
Université Paris Cité

- At the pedagogical commission of the department of mathematics which is in charge of taking decisions regarding everything that is being taught at the university.

Masters-Phd students meeting 2022 and 2023

- Organising an afternoon of short talks followed by an informal discussions between phd students and masters students

Student representative

September 2017 - May 2018

Sorbonne Université

- Teaching Board of the Department of physics of Sorbonne Université
- Keeping track of inovative teaching technics and methods. Measuring their efficiency.

TALKS AND POSTERS

In Conferences

- Upcoming : *Pure minimal injective resolutions and perfect modules for lattices* (15 minutes), ARTIG 8 (Algebra and Representation Theory In Germany), Bielefeld, May 2026.
- *Fractionally Calabi-Yau Posets: coroborating a conjecture by Chapoton* CHARMS summer school and conference in May 2024, Bonn in October 2024, Colloque tournant du RT algèbre in March 2025.
- *A small dictionary for the representation theory of lattices*, Artig 5, Bonn, February 2025
- *Antichains in the representation theorie of Lattices*, Poster, FPSAC 2024.

Invitations to seminars

- Upcoming : *Boolean antichains and Perfect modules*, Osnabrück, April 2026.
- Upcoming : *Pure minimal injective resolutions and perfect modules*, Amien, April 2026.
- *Counting Boolean antichains*, Bonn, October 2025.
- *Combinatorial perspectives on Fractionally Calabi-Yau lattices*, Versailles, Seminaire et groupe de travail d'Algèbre-Géométrie, September 2025.
- *Fractionally Calabi-Yau Lattices: techniques and perspectives* in Strasbourg, Algebra seminar, May 2025. Also in the online seminar ACPMS in June 2025.
- *Fractionally Calabi-Yau Posets: coroborating a conjecture by Chapoton*, GATo seminar, LAMFA, Amien, January 2024.
- *Some homology with antichains in a lattice*, DGeCo seminar, LAMFA, January 2023.
- *Representations of ordered sets and derived equivalences*, LAMFA's PhD student seminar, Amien, November 2022.

Learning seminars

- *Lexicographically shellable posets*, Learning seminar on Cohen-Macaulay Posets, Winter semester 2025-2026, Bochum.
- *Proof of the finitistic conjecture for special algebras*, Learning seminar on the finitistic dimension conjecture, Summer semester 2025, Bochum.
- *6-functor formalism*, Learning seminar on Perverse sheaves, 2021, Université Paris Cité.

CONFERENCES AND SUMMER SCHOOL

- Upcoming : ICRA July 2026
Grenoble
- Upcoming : ARTIG 8 May 2026
Bielefeld

· <u>ARTIG 7</u> <i>Munich</i>	December 2025
· <u>Collaborations in Algebra, Rep. Theory and Ethics (CARE)</u> <i>Lyon</i>	October 2025
· <u>ARTIG 6</u> <i>Bochum</i>	July 2025
· <u>Frieze patterns in algebra, combinatorics and geometry</u> <i>CIRM - Luminy</i>	May 2025
· <u>Spring school on Dynkin Classification</u> <i>Bochum</i>	April 2025
· <u>Colloque tournant du RT algèbre</u> <i>Clermont-Ferrand</i>	March 2025
· <u>ARTIG 5:</u> <i>Bonn</i>	February 2025
· <u>ABCD meeting</u> University of Cologne	October 2024
· <u>FPSAC</u> <i>Bochum</i>	July 2024
· <u>Rencontre de l'ANR CHARMS: Catégories amassés et symétrie miroir</u> <i>Strasbourg</i>	May 2024
· <u>CHARMS Summer School</u> <i>Versailles</i>	May 2024
· <u>Current trends in Rep. theory, Cluster algebras and Geometry</u> <i>CIRM - Luminy</i>	November 2023
· <u>LMS-Bath Summer school on Geometric and Categorical Rep. theory</u> <i>Bath</i>	August 2023
· <u>Homology of groups and functors</u> <i>Université de Lille</i>	June 2023
· <u>Algebra days in Paris in honor of Bernhard Keller</u> <i>Institut Henri Poincaré - Paris</i>	September 2022
· <u>Summer school Representation theory and flag or quiver varieties</u> <i>Université Paris Cité</i>	June 2022
· <u>Summer school in Symplectic singularities in Geometry and Rep. theory</u> <i>CIRM - Luminy</i>	April 2022
· <u>Summer School in Algebraic Combinatorics</u> <i>online</i>	July 2021

RESEARCH VISITS

- Visiting Viktória Klász, Markus Kleinau, Rene Marczinzik in Bonn in October 2025
- Junior Research Retreat in Bad Boll: Representation Theory and Combinatorics in September 2025, working with Julia Hörmayer, Maximilian Kaipel and Viktória Klász.
- Visiting Viktória Klász, Markus Kleinau, Rene Marczinzik in Bonn in October 2024

TEACHING

Ruhr-Universität-Bochum

October 2024 - Present

- Category theory. Winter semester 2025-2026. I taught the second half of the course which consisted in two session of one and half hours of lectures combined with exercises. The first half of the course introduced the language of category theory and followed the book *Categories in Context* by Emily Riehl. The second half of the class explored more advanced topics such as 2-dimensional TQFTS on request of the students. **Course coordinators:** Dr. Azzurra Ciliberti and Dr. Tal Gottesman.
- Elementary number theory. Summer semester 2025 and 2026. I conducted weekly one and half hours long exercise classes for a group of ten second year undergraduate students from the bachelor of arts of the University of Bochum. In 2026 I plan on giving the example class in german **Course coordinators:** Prof. Dr. Markus Reineke and Dr. Nico Lorenz; Prof. Dr. Alex Ivanov and Felix Zillinger.
- Representation theory of quivers. Winter semester 2024-2025. I conducted weekly one and a half hours long exercise classes for a group of four masters student at the university of Bochum. I prepared the exercise sheets. **Course coordinator:** Prof. Dr. Karin Baur.

Université Paris Cité

September 2021 - August 2024

Complementary teaching duty as a phd student

- Integration and Fourier series. I conducted examples classes for a group of 35 third year undergraduate students in mathematics for three hours every week. **Course coordinator:** Cyril Lucas.
- Algorithms and Complexity for first year graduate students in mathematics. I conducted practice sessions for 20 students for four hours everyweek for a month then two hours every two weeks for a month. The end of a the course consisted in supervising the student's end of term programming project. **Course coordinators:** Bertrand Gentou, François Le Maitre.
- Oral exams ("colles") in algebra and analysis for first year undergraduate students in mathematics. Every week I would prepare a series of exercises related to the classes the student were taking. I would then use those exercises to simultaneously evaluate three students each solving a different exercise on a black board for an hour. **Course coordinator:** Marie-Claude Arnaud, Patrick Le Meur.

AWARDS AND STUDENTSHIPS

Programme PGSM

2019/2021

Fondation Science Mathématique de Paris

- Full studentship

Knox Studentship

2018/2019

Trinity College - Cambridge

- Full studentship

Award for excellency

June 2016

Sorbonne Université

- 600 € for finishing 3rd out of 800 students during the first year of undergraduate studies

WORKING PAPERS

About des representations of preordered sets

June 2020

Introduction to research

- Under the supervision of Sebastian Posur (University of Siegen), Baptiste Rognerud (University of Paris) and Daniel Juteau (University of Paris).
- I reproduce a proof of the existence of an equivalence between a finite poset category of representations and the Freyd Category of the additive closure of the linear closure of its associated category. Using this equivalence I was able to implement the derived category of a finite poset set in the Gap Package CAP. The intention was to identify interesting derived equivalences conjectured by Chapoton in Lie theory.

Congruent Subgroup Problems

June 2019

Part III Essay

- Under the supervision of Emamnuel Breuillard.
- Reading the article "Solution to the congruence subgroup problem" de Bass, Milnor et Serre.

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB, GAP, SageMath
Software & Tools	LaTeX, Excel, Mathematica, GAP
Languages	French, Hebrew (mother tong), English (TOEFL) German (B1)